Black-Box System Investigation

Analyze an unknown input-output system and develop a functional model without opening it.

Intermediate	1-2 class periods	Engineering Challenge

Engineering Context

Mission engineers frequently work with subsystems supplied by another team. They may know the inputs and outputs but not every internal detail. Your team must characterize an unknown classroom system.

What You Will Practice

- Define system boundaries
- Identify inputs, processes, outputs, and feedback
- Plan controlled tests
- Create and revise a functional model

What You Need

- Instructor-provided black-box device or simulation
- Approved test inputs
- Stopwatch or measurement tools
- Engineering notebook

Student Instructions

1. Observe the system without operating it and define the system boundary.
2. List possible inputs, outputs, energy transfers, and internal processes.
3. Plan at least four controlled tests that change one input at a time.
4. Run the tests and record quantitative and qualitative outputs.
5. Create an initial block diagram.
6. Use evidence to revise the model and predict a new result.
7. Run one confirmation test and evaluate the prediction.

Engineering Requirements

- At least four controlled tests
- At least one quantitative output
- Functional model must show input, process, output, and feedback if present
- Final prediction must be tested

Constraints & Safety

- Do not open, alter, overload, or bypass safety limits on the system
- Use only approved inputs
- Stop testing if behavior becomes unsafe or inconsistent

What You Submit

- Test plan
- Data table
- Initial and revised system diagrams
- Prediction and confirmation result
- Reflection

Reflection Questions

1. Which test provided the most useful information?
2. What part of the internal process remains uncertain?
3. How did your model change after testing?
4. What additional sensor would improve the investigation?

Engineering Tip Choose tests that isolate one variable rather than changing several conditions at once.	**Common Mistake** Assuming the first explanation is correct can lead to confirmation bias. Actively look for evidence that could disprove it.
Stretch Goal Create a fault scenario and explain how the outputs would change.	**Career Connection** Systems engineers use functional models to integrate propulsion, avionics, structures, and controls developed by different teams.